DepoRank

Term deposit campaign prioritization

Five-model comparison and a frozen contact-ranking application

COMP6140001 Data Mining
BINUS University | Academic Year 2026/2027

A fixed quota needs ranking, not a yes label

Retrospective test evidence for the selection-frozen GaussianNB model

824

contact priorities

512

historical yes outcomes

2.02

lift over test prevalence

62.14% Precision@K | 20.16% Recall@K | 8,238 test observations

Historical outcomes are not causal uplift or a forecast of future subscriptions.

Campaign planning within limited capacity

Stakeholder: a campaign analyst allocating a fixed calling budget

Business question

Which contact observations should receive the available campaign slots?

Application output

A validated score, a fixed-threshold label, and exactly K contact priorities.

Measure retrieval and capacity utility. Do not claim ROI or incremental response.

The dataset is historical and imbalanced

UCI bank-additional-full | Portuguese bank contacts, May 2008-November 2010

41,188

contact observations

20 inputs + binary target y
12 duplicates retained
Unknown categories preserved



yes 11.27% | no 88.73%

KDD turns raw records into auditable decisions

All five required stages are present in the application and report

1

Selection

Official full CSV
Core attributes
Checksum pin

2

Preprocessing

Validate schema
Audit duplicates
Reject bad rows

3

Transformation

pdays sentinel
One-hot encoding
Train-only fit

4

Data mining

Five families
Temporal CV
Calibration

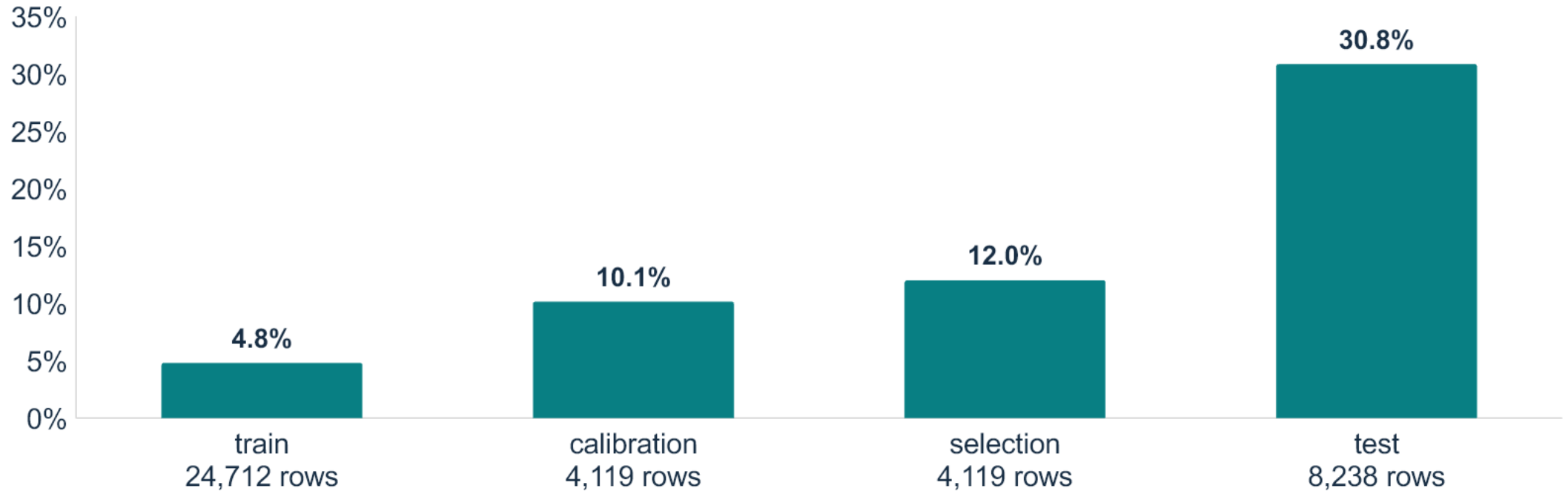
5

Evaluation

Selection lock
Final test
Capacity metrics

Prevalence changes across four ordered blocks

Original CSV positions preserved | No shuffling or refitting on future blocks



Train: fit/search | Calibration: probability mapping | Selection: decisions | Test: final evidence

Five distinct model families share one protocol

Same raw core features, chronological blocks, and evaluation definitions

Model family	Mechanism	Key tradeoff
Logistic Regression	Linear log-odds	Compact; limited nonlinear interactions
Gaussian Naive Bayes	Class-conditional likelihoods	Fast; conditional independence assumption
Decision Tree	Recursive rules	Nonlinear; unstable when overfit
Random Forest	Averaged randomized trees	Lower variance; higher memory cost
Gradient Boosting	Sequential additive trees	Flexible; tuning and drift sensitivity

Dummy prior and random ranking are diagnostic baselines, not extra learned families.

Keep training, calibration, and selection separate

10 core features | 41 candidates | 3 expanding-window folds | 123 fold fits

Leakage controls

Exclude duration and campaign
Never fit preprocessing on future data
Handle pdays=999 as a sentinel
Keep row IDs out of the model

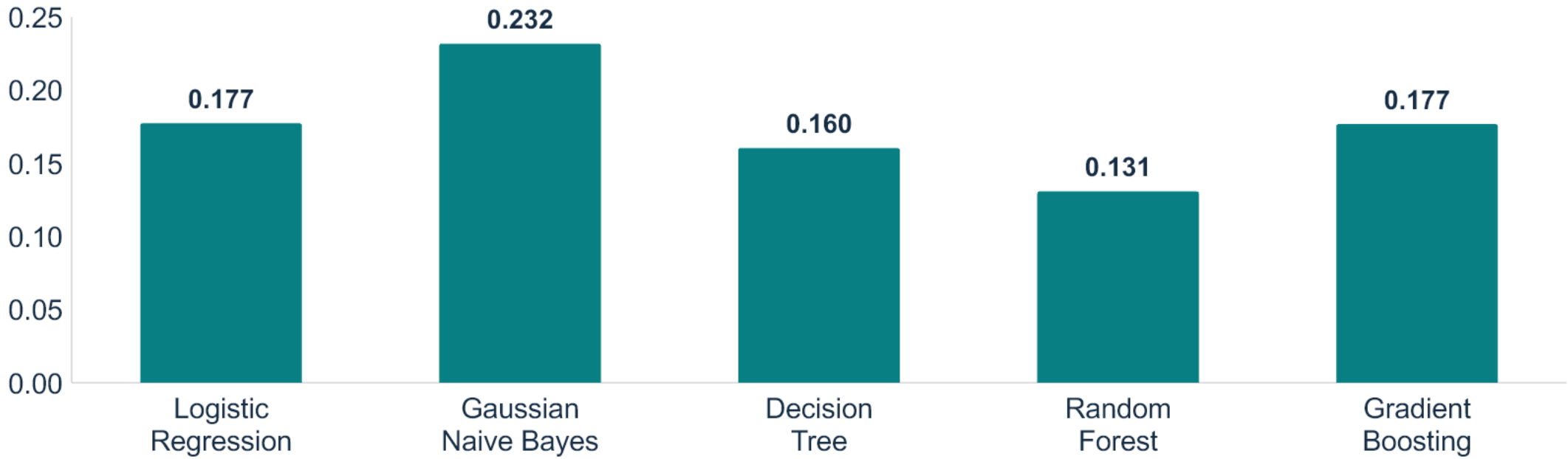
Decision controls

FrozenEstimator sigmoid calibration
Selection AP chooses the winner
Selection F1 chooses thresholds
Final test cannot select a new winner

A threshold label does not determine the top-K contact quota.

GaussianNB wins before test results are opened

Selection AP is the declared primary metric | GaussianNB threshold fixed at 0.09



The selection lock pins the winner, thresholds, models, and dataset/config hashes.

Final test results show different tradeoffs

8,238 observations | K=824 | Thresholds fixed on selection | No model reselection

Model	AP	ROC-AUC	P@K	R@K	Lift@K	Brier
Logistic Regression	0.4130	0.5364	0.6517	0.2114	2.1137	0.2455
Gaussian Naive Bayes	0.5076	0.6667	0.6214	0.2016	2.0153	0.2546
Decision Tree	0.3893	0.5531	0.4879	0.1583	1.5823	0.2523
Random Forest	0.3415	0.5769	0.2852	0.0925	0.9250	0.2558
Gradient Boosting	0.4012	0.5363	0.5510	0.1787	1.7870	0.2419

GaussianNB remains active even though Logistic Regression has higher test Precision@K.

Capacity and classification answer different questions

A 10% contact quota is not the same as a probability threshold

Contact quota

824 selected for contact
512 yes within capacity
312 no within capacity
2,028 yes outside capacity

Frozen threshold 0.09

5,965 predicted yes labels
34.53% precision
81.10% recall
46.77% accuracy

No costs, deposit values, consent, or randomized no-contact outcomes are available.

Good ranking does not guarantee reliable probabilities

Test Brier 0.2546 | Large prevalence shift | Production transfer is unproven

4.81%

train yes prevalence

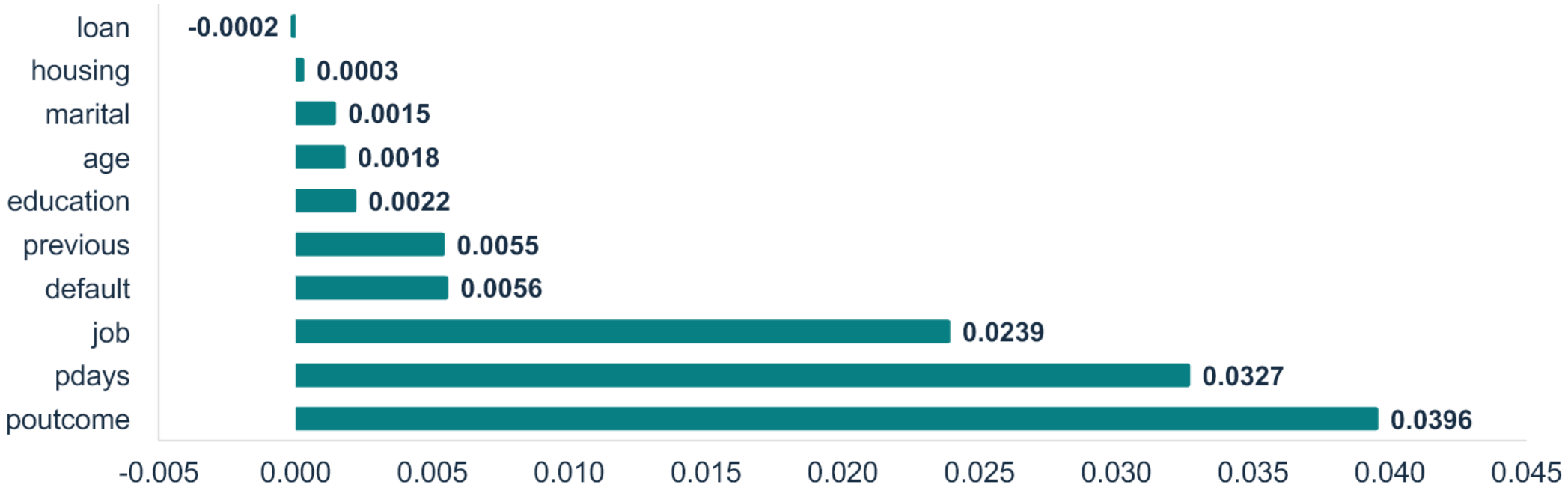
30.83%

test yes prevalence

Validate calibration on new data. Monitor drift, label delay, and capacity performance.

Previous campaign history matters on selection

Permutation importance | Five repeats | Association, not causal explanation



Mean selection AP decrease; loan retains its negative importance (-0.0002).

Correlated features and small subgroups limit attribution. No fairness guarantee is made.

The application demonstrates input through export

Local scoring with a trusted frozen model | No training during page reruns

01 Inspect

KDD and model evidence

02 Predict

One synthetic observation

03 Score

Valid CSV at q=10%

04 Rerank

Change K without training

05 Export

CSV/JSON and provenance

06 Reject

Invalid batch and error report

Software quality: 244 passing tests; predictive results remain separate.

A next experiment must use current, permissioned data

Historical Portugal results do not establish production validity in Indonesia

Current limitations

Old and geographically different data
No verified customer/time isolation
No ROI or treatment-effect evidence
Calibration and fairness unproven

Next validation

Point-in-time customer snapshots
New out-of-time holdout
Drift and subgroup monitoring
Prospective controlled campaign

Preserve the current frozen evidence. New decisions require a new evaluation protocol.

Approve a validation study, not a production launch

Retrospective ranking is promising; future probability reliability remains unproven

Measured on the frozen test

62.14% Precision@824
20.16% Recall@824
2.02 lift over test prevalence
46.77% threshold accuracy

Decision requested

Approve access to current data
Define capacity and contact eligibility
Test calibration before probability claims
Measure response in a controlled study

Reproducible application and evidence: github.com/DlegendZ/DepoRank